Airbnb Market Intelligence – Project Log (Track A: Data Analytics)

AnalystLab Africa Experience Lab, Week 1. Running log of every step, decision and result. This becomes the basis for the README and the data quality report.

Environment

- Database engine: Microsoft SQL Server (managed in SSMS)
- Database: AirbnbMarket
- Reporting: Power BI Desktop (connects directly to SQL Server)
- Excel: not used (per assignment no-Excel policy)

Task 1 – Understanding the business problem

The business problem. Airbnb listings are spread across 10 cities, and each city has its own currency, property mix, pricing and host behaviour. Stakeholders currently have no single, validated view of how listings are distributed, what drives advertised prices, how hosts perform, or how review activity varies. This project builds an interactive Power BI dashboard that answers those questions from the supplied listing and review records.

Why it matters. Short-term rental markets are competitive and differ sharply between cities. Without structured analysis, hosts, investors and platform teams rely on guesswork when judging where supply is concentrated, which property types command higher advertised prices, and where host service standards lag. Reliable, like-for-like market evidence reduces that guesswork.

Intended users.

- Market and strategy analysts comparing city markets.
- Property investors and hosts assessing where and what type of property to list.
- Platform operations teams monitoring host responsiveness and listing quality.

Decisions the dashboard can support.

- Which cities and neighbourhoods are saturated or under-supplied with listings.
- How to position a listing's advertised price against comparable room types and capacity within the same city.

- Which cities need host-support programmes to improve response and acceptance rates.
- Which property segments attract the most review activity and the strongest listing-level scores.

Limitations of the evidence.

- Prices are advertised amounts in each city's local currency, with no currency field, so raw prices are compared only within a city.
- Advertised price is not actual revenue or profit. The data has no bookings, completed stays or occupancy.
- Review records contain only dates and IDs, with no written comments, so sentiment cannot be analysed. Review count indicates activity, not satisfaction.
- Key fields such as district, bedrooms, host response and acceptance rates and review scores have missing values.
- 160 review IDs are repeated and must be investigated before review_id is treated as unique.
- Findings show associations, not causes.

Step 1 – Data import

Method: SSMS Import Flat File wizard (Database → Tasks → Import Flat File).

Decision: No primary key was set during import. Reason: listing_id uniqueness must be verified first (assignment section 4.3). The key will be added after validation.

Issues encountered and fixes:

1. Error: *String or binary data would be truncated*, column `name`. The wizard sized text columns from a sample of rows; some listing names are longer. Fix: widened text columns to nvarchar(MAX).
2. Error: value `1001` could not be converted to tinyint, column `minimum_nights`. The wizard guessed tinyint (max 255) but at least one listing has minimum_nights = 1001. This is recorded as an unusual value to investigate.

Final approach: All columns imported as nvarchar(MAX) with nulls allowed (raw staging load). Data types will be converted in SQL using TRY_CAST so that invalid values can be found and documented rather than blocking the import.

Row count check: Listings expected 279,712 – result 279,712 (matches). Reviews expected 5,373,143 – result 5,373,143 (matches). Both tables imported completely.

Step 2 – Data inspection

2.1 Listings primary key: `SELECT COUNT(*) AS total_rows, COUNT(DISTINCT listing_id) AS unique_ids FROM Listings;` returned 279,712 total rows and 279,712 unique IDs. listing_id is unique and valid as the primary key.

2.2 Repeated review IDs: A subquery grouped Reviews by review_id and kept IDs with COUNT(*) > 1. The outer query counted them. Result: 160 repeated review_id values, matching the assignment brief. review_id alone is therefore not a unique key. Next: inspect these rows and test whether listing_id + review_id is unique.

2.3 Inspecting the repeated rows: Selected all Reviews rows whose review_id is in the repeated list, ordered by review_id. Result: 320 rows (160 IDs × 2), so each repeated ID appears exactly twice. In each pair the review_id, date and reviewer_id are the same, but the listing_id is different. Interpretation: the same review record is linked to two different listings, likely a source-data artefact such as a relisted or duplicated property. These are not exact duplicate rows.

Treatment decision: Keep all 320 rows. Each row is a valid listing–review link, and deleting one of each pair would arbitrarily remove a review from a listing. Use listing_id + review_id as the composite key. Review counts per listing include both. Total review records = 5,373,143; distinct review IDs = 5,373,143 − 160 = 5,372,983. Both figures are reported and labelled on the dashboard.

2.4 Composite key check: Counted all rows and distinct listing_id + review_id pairs (joined with CONCAT and a separator). Result: 5,373,143 total and 5,373,143 unique pairs. The composite key listing_id + review_id is confirmed unique.

2.5 Orphan review check: LEFT JOIN Reviews to Listings on listing_id, counting reviews with no matching listing. Result: 0 orphan reviews. Every review links to an existing listing, so the one-to-many relationship (one listing, many reviews) is valid and no reviews will drop out when the tables are joined in Power BI.

2.6 Listings without reviews: For each listing, NOT EXISTS checked whether any review carries its listing_id, and counted those with none. Result: 86,156 listings (about 30.8% of 279,712) have no review records. They are kept in all listing counts. Review-based measures (reviews per listing, review activity) must state whether they include or exclude these listings.

2.7 Missing values (of 279,712 listings): NULLIF treated empty text as blank, CASE scored each blank as 1, and SUM added them per column.

Column	Missing	Share
district	242,700	86.8%
host_response_rate	128,782	46.0%
host_acceptance_rate	113,087	40.4%
review_scores_rating	91,405	32.7%
bedrooms	29,435	10.5%

Interpretation: district is too incomplete for cross-city district analysis, so neighbourhood will be used for location concentration. Missing ratings (91,405) are close to the 86,156 listings with no reviews, which is expected: no reviews means no score. Treatment: missing values are left blank, not filled with zero or averages, so averages are calculated only on listings that have the value, and the dashboard shows how many listings each measure is based on.

2.8 Zero or negative prices: Converted price from text to a number with TRY_CAST, kept listings with price <= 0, and counted them per city with GROUP BY. Query written by me. Result: 113 listings in 7 cities.

City	Listings with price <= 0
Paris	62
New York	28
Rio de Janeiro	7
Mexico City	7
Rome	6
Istanbul	2
Bangkok	1

The remaining cities had none. These prices are invalid for pricing analysis (a nightly price cannot be zero) and will be excluded from price measures only. The listings stay in listing counts.

2.9 Blank or non-numeric prices: Kept listings where TRY_CAST(price AS DECIMAL(18,2)) returns NULL (meaning the price is blank or cannot be read as a number) and counted them per city. Query written by me. Result: 0 rows. Every listing has a price that converts to a valid number, so price can safely be converted to DECIMAL at the cleaning step.

2.10 Price range per city: MIN and MAX of the converted price, grouped by city. This confirms all 10 cities are present. Prices are in each city's local currency and are not compared across cities.

City	Lowest price	Highest price
Rio de Janeiro	0.00	625,216.00
Mexico City	0.00	499,000.00
Bangkok	0.00	300,177.00
Cape Town	122.00	180,000.00
Istanbul	0.00	179,532.00
Hong Kong	1.00	83,665.00
Sydney	5.00	28,613.00
Paris	0.00	12,000.00
Rome	0.00	10,571.00
New York	0.00	10,000.00

Interpretation: every city has extreme highs far above a normal nightly price (for example 12,000 EUR a night in Paris), and some suspicious lows above zero (1 HKD in Hong Kong, 5 AUD in Sydney). Extremes may be luxury properties or data-entry placeholders.
Treatment: zero or negative prices are excluded from price measures. Extreme highs are kept but flagged, and the median is used instead of the average so a few extremes cannot distort typical prices.

2.11 Value formats (data type check): Sampled rows with SELECT TOP 10 to see how values are stored before converting types.

Column	Stored as	Converts to
host_since	2011-12-03 (year-month-day text)	DATE
host_response_rate, host_acceptance_rate	decimals 0 to 1 (e.g. 0.9, 0.67)	DECIMAL; shown as % on the dashboard
host_is_superhost, instant_bookable	t / f	1 / 0
bedrooms, accommodates	whole numbers	INT
review_scores_rating	0 to 100 scale (e.g. 100)	DECIMAL
price	numeric text, local currency	DECIMAL(18,2)

Pending: duplicate review_id investigation, listing–review relationship, missing values, data types, invalid prices.

Step 3 – Cleaning and data model

3.1 Listings_clean: Created with SELECT … INTO from the raw Listings table, converting each column with TRY_CAST (IDs to BIGINT, host_since to DATE, rates and ratings to DECIMAL, counts to INT, price to DECIMAL(18,2)) and t/f flags to 1/0 with CASE. Added a price_valid flag (1 when price > 0, else 0). The raw table is left unchanged. Checks: 279,712 rows (matches raw); SUM(price_valid) = 279,599 (279,712 minus the 113 zero or negative prices).

3.2 Reviews_clean: Created the same way: listing_id, review_id and reviewer_id to BIGINT, date to DATE (renamed review_date; square brackets used because date is a reserved SQL word). Check: 5,373,143 rows (matches raw).

3.3 Reviews per listing: Grouped Reviews_clean by listing_id and counted rows with COUNT(*). Query written by me. Predicted rows: 279,712 listings minus 86,156 with no reviews = 193,556. Result: 193,556 rows, matching the prediction and confirming the earlier no-review check.

3.4 View vw_reviews_per_listing: Saved the reviews-per-listing query as a view (a stored query that recalculates from Reviews_clean each time it is read), with columns listing_id and Reviews_count. Power BI loads this small summary instead of 5.3 million review rows. Tested with SELECT TOP 10: returns one row per listing with its review count.

3.5 View vw_reviews_per_month_city: JOINed Reviews_clean to Listings_clean on listing_id to attach each review's city, rounded each review_date to the first of its month with DATEFROMPARTS(YEAR, MONTH, 1), and counted reviews per city per month. This feeds the review-activity-over-time chart. Check: SUM(review_count) across the view = 5,373,143, matching total review records, so no reviews were lost in the join or grouping. Check query written by me.

3.6 Power BI connection: Power BI Desktop connected to SQL Server (BOMOTAYO\SQLEXPRESS, database AirbnbMarket) in Import mode. Loaded only Listings_clean, vw_reviews_per_listing and vw_reviews_per_month_city. The raw tables and the 5.3 million review rows stay in SQL Server, so heavy processing happens in SQL as the brief requires. The first load took about 30 minutes because Power BI runs each view's query (a join and aggregation over 5.3M rows) on refresh. If refresh is slow, the views can be stored as tables with SELECT * INTO.

3.7 Data model (Track A Task 1, point 8):

- Listings_clean (1) – (1) vw_reviews_per_listing on listing_id: each listing matched to its review count.
- A new Cities table, created with the DAX formula Cities = DISTINCT(Listings_clean[city]), holds the 10 cities.
- Cities (1) → (many) Listings_clean on city.
- Cities (1) → (many) vw_reviews_per_month_city on city.

Cities is the shared lookup table, so one city slicer filters listing, pricing, host and review-activity visuals together. The monthly view has no listing_id, so it links through Cities rather than directly to Listings_clean.

Track A Task 2 – Business questions

At least six questions are required. Q1 (listings per city), Q7 (reviews per listing) and Q8 (review activity over time) are already covered by queries and views above; results to be recorded here.

Q1 – Which cities have the highest number of listings?

- KPI: Total Listings by city.
- Calculation: COUNT(*) grouped by city in SQL; Total Listings = COUNTROWS(Listings_clean) by Cities[city] in Power BI (dashboard page 1).

City	Listings	Share
Paris	64,690	23.1%
New York	37,012	13.2%
Sydney	33,630	12.0%
Rome	27,647	9.9%
Rio de Janeiro	26,615	9.5%
Istanbul	24,519	8.8%
Mexico City	20,065	7.2%
Bangkok	19,361	6.9%
Cape Town	19,086	6.8%
Hong Kong	7,087	2.5%

The ten cities add up to 279,712, matching total listings.

- Business interpretation: Paris is by far the largest market, with almost one in four listings and nearly twice New York's count. Hong Kong is the smallest at 2.5%. Supply is concentrated in a few cities, so overall figures lean heavily on Paris; city-level views are needed to avoid Paris dominating the totals.

Q2 – How are listings distributed across room types?

- KPI: listing count and share of listings by room type.
- Calculation: COUNT(*) on Listings_clean grouped by room_type, ordered by count DESC. Query written by me.

Room type	Listings	Share
Entire place	182,005	65.1%
Private room	86,988	31.1%

Room type	Listings	Share
Hotel room	5,857	2.1%
Shared room	4,862	1.7%

The four counts add up to 279,712, matching total listings.

- Business interpretation: about two in three listings are entire homes, so the marketplace is dominated by whole-property rentals rather than shared hosting. Private rooms are a substantial second segment (about a third). Hotel and shared rooms are niche at under 4% combined, so pricing benchmarks for those segments rest on small samples.

Q3 – Which property types are most common?

- KPI: listing count and share of listings by property type (top 10).
- Calculation: SELECT TOP 10 property_type, COUNT(*) on Listings_clean, grouped by property_type, ordered by count DESC. Query written by me.

Property type	Listings	Share
Entire apartment	138,989	49.7%
Private room in apartment	47,322	16.9%
Private room in house	13,292	4.8%
Entire house	13,273	4.7%
Entire condominium	11,250	4.0%
Room in boutique hotel	5,771	2.1%
Entire loft	4,587	1.6%
Private room in condominium	4,462	1.6%
Private room in bed and breakfast	4,238	1.5%
Entire serviced apartment	3,973	1.4%

The top 10 cover 247,157 listings (88.4%); the remaining types make up a long tail.

- Business interpretation: this is an apartment-led market. Entire apartments alone are half of all listings, and apartments (entire or a private room in one) make up two thirds. Houses, condos and boutique-hotel rooms are much smaller segments. Hosts listing an entire apartment face the most competition, while less common types may stand out more.

Q9 – How do host response and acceptance indicators vary across cities?

- KPI: average host response rate and average host acceptance rate by city.
- Calculation: AVG(host_response_rate) and AVG(host_acceptance_rate) on Listings_clean, grouped by city. AVG ignores blank values, so the averages use only listings that report a rate (46.0% of listings have no response rate and 40.4% no acceptance rate). Query written by me.

City	Avg response rate	Avg acceptance rate
Cape Town	90.8%	79.7%
Mexico City	89.3%	88.9%
New York	88.5%	80.5%
Sydney	87.9%	78.6%
Rome	87.5%	89.9%
Paris	85.9%	80.7%
Bangkok	85.1%	90.5%
Rio de Janeiro	83.5%	75.7%
Hong Kong	83.3%	80.1%
Istanbul	83.0%	85.2%

- Business interpretation: response rates are fairly consistent, from 83.0% (Istanbul) to 90.8% (Cape Town). Acceptance varies more, from 75.7% (Rio de Janeiro) to 90.5% (Bangkok). Rio de Janeiro and Hong Kong are below average on both, so they are the first candidates for host-support programmes. Because so many listings do not report these rates, the averages describe reporting hosts only.

Q9 (continued) – Other host characteristics by city (dashboard page 3)

City	Superhost share	Listings per host
Mexico City	32%	1.7
Rome	26%	1.9
Cape Town	24%	1.7
Bangkok	20%	2.3
New York	19%	1.4
Hong Kong	18%	3.5
Rio de Janeiro	17%	1.5
Istanbul	13%	1.8
Paris	13%	1.2
Sydney	12%	1.3

- Business interpretation: Mexico City has the strongest host quality signal (almost one in three listings run by a superhost). Hong Kong is dominated by multi-property hosts (3.5 listings per host), which suggests professional operators rather than home-sharers. Paris is the largest market but has among the lowest superhost share (13%) and the fewest listings per host (1.2), so it is mostly individual hosts renting their own homes.

Q4 – How do advertised prices vary across cities, within each local currency?

- KPI: median advertised nightly price per city, in local currency; the average is shown only for comparison.
- Calculation: PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY price) OVER (PARTITION BY city) for the median and AVG(price) OVER (PARTITION BY city) for the average, on Listings_clean WHERE price_valid = 1 (excludes the 113 zero or negative prices). DISTINCT returns one row per city.
- Currency: the data has no currency field. Currencies are assumed from each city's local currency and should be labelled as such on the dashboard.

City	Currency (assumed)	Median price	Average price	Average ÷ median
Rio de Janeiro	BRL	280	742.78	2.65

City	Currency (assumed)	Median price	Average price	Average ÷ median
Cape Town	ZAR	1,069	2,405.12	2.25
Istanbul	TRY	252	532.60	2.11
Hong Kong	HKD	386	746.17	1.93
Bangkok	THB	1,100	2,078.39	1.89
Sydney	AUD	120	222.01	1.85
Mexico City	MXN	664	1,149.65	1.73
Rome	EUR	65	105.13	1.62
New York	USD	99	142.95	1.44
Paris	EUR	80	113.20	1.42

- Business interpretation: in every city the average is well above the median (1.4 to 2.7 times), confirming that a small number of very high prices skew averages. The median is therefore the right measure of a typical price. Skew is strongest in Rio de Janeiro and Cape Town, so price benchmarks there are most sensitive to outliers. Raw prices are not compared across currencies. The only like-for-like comparison is Paris against Rome (both EUR), where the typical Paris listing (80) is about 23% dearer than Rome (65).

Q7 – How many reviews are recorded for each listing (and city)?

- KPI: Total Review Records and review records per listing, by city.
- Calculation: vw_reviews_per_listing counts reviews per listing in SQL; on dashboard page 4, Total Review Records by Cities[city]. Reviews per listing = city review records ÷ city listings.

City	Review records	Listings	Reviews per listing
Rome	1,113,360	27,647	40.3
Mexico City	477,584	20,065	23.8
New York	847,727	37,012	22.9
Hong Kong	134,228	7,087	18.9

City	Review records	Listings	Reviews per listing
Paris	1,213,727	64,690	18.8
Cape Town	302,336	19,086	15.8
Bangkok	284,342	19,361	14.7
Sydney	482,552	33,630	14.3
Rio de Janeiro	323,274	26,615	12.1
Istanbul	194,013	24,519	7.9

Overall: 19.2 review records per listing; 86,156 listings (30.8%) have none.

- Business interpretation: Paris has the most review records in total, but Rome is the most active per listing (40.3), more than twice Paris and five times Istanbul (7.9). Review activity is a proxy for guest demand rather than a measure of bookings. Rome listings appear to turn over guests far more often, while Istanbul and Rio have large supply with comparatively little review activity.

Review count vs rating (scatter, page 4): ratings cluster between 80 and 100 for listings with many reviews; low ratings appear only among listings with few reviews. Busier listings are consistently well rated, an association rather than proof of cause.

Q8 – How does review activity change over time?

- KPI: Total Review Records by year (from vw_reviews_per_month_city).
- Calculation: SUM(review_count) by review_month, shown per year on dashboard page 4. Date range checked in SQL with MIN and MAX of review_date: 16 Nov 2008 to 1 Mar 2021. The yearly values sum to exactly 5,373,143.

Year	Review records
2008	2
2009	115
2010	1,236
2011	6,253
2012	19,922
2013	50,522

Year	Review records
2014	122,132
2015	280,332
2016	501,754
2017	777,200
2018	1,142,496
2019	1,633,546
2020	755,324
2021 (to 1 Mar only)	82,309

- Business interpretation: review activity grew every year from 2008 to a peak of 1.63 million in 2019, then fell 54% in 2020, consistent with COVID-19 travel restrictions. 2021 is a partial year (data ends 1 March 2021), so its low figure is not a further collapse and is labelled on the dashboard. Month-of-year view: October is the busiest month (552,877 reviews across all years) and April the quietest (375,740).

Step 4 – KPI definitions

All KPIs are DAX measures stored in Listings_clean, so every visual reuses one definition. Values shown are for all 10 cities with no filters applied.

KPI	DAX formula	Definition and interpretation	Value
Total Listings	COUNTROWS(Listings_clean)	Number of listings. Measures market supply.	279,712
Number of Cities	DISTINCTCOUNT(Listings_clean[city])	Distinct cities in the current filter.	10

KPI	DAX formula	Definition and interpretation	Value
Median Price	CALCULATE(MEDIAN(Listings_clean[price]), Listings_clean[price_valid] = 1)	Middle advertised nightly price, excluding zero or negative prices. Local currency only, so it is read per city and never across cities.	Per city (see Q4)
Total Review Records	SUM(vw_reviews_per_month_city[review_count])	All review rows, including the 160 repeated review IDs on different listings. Indicates review activity, not satisfaction.	5,373,143 (matches SQL)
Avg Reviews per Listing	DIVIDE(SUM(vw_reviews_per_listing[Reviews_count]), [Total Listings])	Review records divided by all listings. Listings with no reviews count as zero.	19.2
Avg Review Score	AVERAGE(Listings_clean[review_scores_rating])	Average listing rating on a 0 to 100 scale, among rated listings only (blanks are skipped).	93.4
Avg Response Rate	AVERAGE(Listings_clean[host_response_rate])	Average share of guest messages hosts respond to, among listings that report it. Shown as %.	86.6%
Avg Acceptance Rate	AVERAGE(Listings_clean[host_acceptance_rate])	Average share of booking requests hosts accept, among listings that report it. Shown as %.	82.7%

KPI	DAX formula	Definition and interpretation	Value
Superhost Share	AVERAGE(Listings_clean[is_superhost])	Share of listings run by a superhost. is_superhost is stored as 1/0, so its average is the share. Shown as %.	18%
Listings per Host	DIVIDE([Total Listings], DISTINCTCOUNT(Listings_clean[host_id]))	Average number of listings per distinct host. Higher values suggest more multi-property (professional) hosts.	1.54

Formatting note: the rates are stored as 0 to 1 and use percentage format. Review score is already out of 100, so it uses decimal format (percentage format would wrongly show 9340.5%).

Step 5 – Dashboard

Power BI report Air_bnb_Market_Dashboard.pbix, four pages on a 1280 × 1000 canvas with the Airbnb Warm theme (imported JSON theme). Every page has a title, a page navigator, City / Room type / Property type slicers, four KPI cards and three charts.

Page	Cards	Charts
Overview	Total Listings, Number of Cities, Total Review Records, Avg Review Score	Listings by city; room type donut; top 10 property types (Top N filter)
Pricing	Median Price, Average Price, Valid Price Listings, Price Skew	Median price by room type; median price by guest capacity; typical price by city table
Hosts	Listings per Host, Superhost Share, Avg Response Rate, Avg Acceptance Rate	Superhost share by city; listings per host by city; response and acceptance by city
Reviews	Total Review Records, Avg Reviews per Listing, Avg Review Score, Listings with No Reviews	Review records by city; review count vs rating scatter; review activity over time

Design decisions: the Pricing city slicer is single select so prices are never mixed across currencies; the Pricing table ignores the city slicer (Edit interactions: None) so it always lists all 10 cities, with totals switched off; the 2021 point on the time chart is labelled as a partial year.

Step 6 – Validation

#	Check	SQL result	Dashboard result	Status
1	Total listings	279,712	279,712 (Page 1 card)	Match
2	Listings per city add up to total	279,712	10 city bars sum to 279,712	Match
3	Room type shares	Entire place 65.1%, private room 31.1%	65.07%, 31.1% (donut)	Match
4	Total review records	5,373,143	5,373,143 (card)	Match
5	Paris median price	80	80.00 (Page 2 card, City = Paris)	Match
6	Paris average price	113.20	113.20 (card)	Match
7	Paris valid-price listings	64,690 − 62 zero prices = 64,628	64,628 (card)	Match
8	Paris price skew	1.42	1.42 (card)	Match

Filter behaviour: selecting Paris in the city slicer updates every card and chart on the page.

Discrepancy found and fixed in design: with the city slicer on All, the Pricing cards showed a median of 150 and a skew of 4.06, because prices in 10 different currencies were being combined. Fix: the Pricing page city slicer is set to single select, so price measures are always read within one currency.

Check 9: all ten cities' prices. The Pricing table (city slicer interaction set to None, so it always lists every city) shows median, average and skew for all 10 cities. All 30 values match the SQL PERCENTILE_CONT and AVG results in Q4 exactly, for example Rio de Janeiro 280 / 742.78 / 2.65 and Paris 80 / 113.20 / 1.42.

Earlier table mismatch: before the interaction change, the table briefly showed Paris at 112.27 against 113.20 on the card and in SQL. After the table was reconfigured it shows 113.20, consistent with every other source; the exact cause of the earlier value was not confirmed.

Table total row: the table's Total row (median 150, average 609.04, skew 4.06) combines all currencies and is meaningless, so totals are switched off on that table.

Check 10: host rates by city (discrepancy found and fixed). The Hosts page chart first showed rates slightly different from SQL (Cape Town response 90.5% against SQL 90.8%; Mexico City 89.9% against 89.3%), while the overall card (86.6%) matched SQL. Diagnosis in SQL: COUNT(host_response_rate) per city sums to 150,930 reporting listings, and 279,712 − 150,930 = 128,782, exactly the missing-rate count from the data quality check. The count-weighted average of the city results is 86.59%, matching the card. Because the total matched but one chart did not, the cause was a filter on that visual: a leftover property_type Top 10 filter copied when Page 1 was duplicated. After removing it, all 20 values (response and acceptance for 10 cities) match SQL exactly. Lesson: check "Filters on this visual" after duplicating pages.

Check 11: reviews page. The cards show 5,373,143 review records, 19.2 reviews per listing, 93.4 average score and 86,156 listings with no reviews, all matching SQL. The 10 city bars in Review records by city sum to exactly 5,373,143.

Step 7 – Findings and recommendations

Written up as a separate deliverable: Executive Summary – Airbnb Market Intelligence, with seven findings, three recommendations (each with evidence, action and evaluation), limitations and a validation summary.

Dashboard layout spec (simple version, every page)

Page size: Power BI default 16:9, 1280 × 720. Set each box under Format › General › Properties › Size and Position.

Box	X (horizontal)	Y (vertical)	Width	Height
Title text box	20	16	640	48
Slicer 1: city (Cities)	688	12	180	56

Box	X (horizontal)	Y (vertical)	Width	Height
Slicer 2: room_type	884	12	180	56
Slicer 3: property_type	1080	12	180	56
Card 1	20	80	300	84
Card 2	333	80	300	84
Card 3	646	80	300	84
Card 4	960	80	300	84
Chart left	20	176	614	236
Chart right	646	176	614	236
Wide chart	20	424	1240	280

Gaps are 12–16 px between every box. Theme colours: #B4471F, #D98E2B, #8C5A3C, #E6C9A8. Slicers use Dropdown style.